

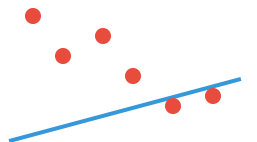
ECO 375

# Tut2: Simple Linear Regression

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Intuition Recap

*Fall 2025*



# What we're doing today

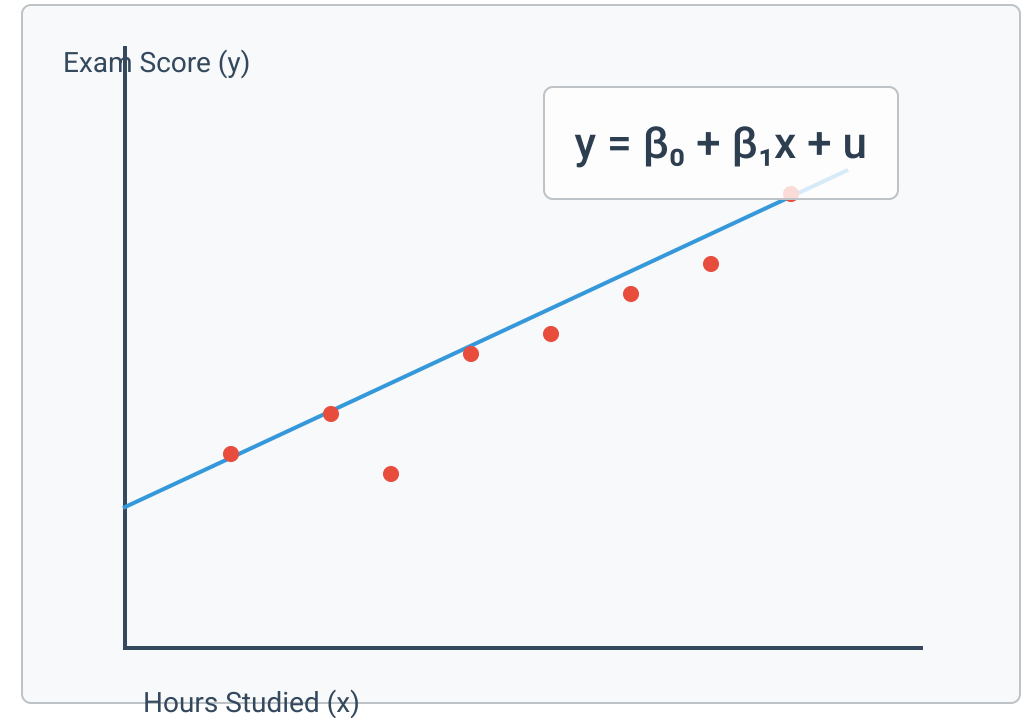
- Big picture → the straight-line idea
- Assumptions that make conclusions believable
- How to test and communicate a slope

## Speaker Notes:

Set expectations: we'll connect the idea of a best-fit line to what we can responsibly claim, and do tiny practices that map to assumptions and inference.

# SLR in one picture

- > We summarize the average relationship:  $y \approx \beta_0 + \beta_1 x$
- >  $\beta_1$  = average change in  $y$  for a 1-unit change in  $x$
- >  $\beta_0$  = baseline when  $x = 0$  (may or may not be meaningful)



## Speaker Notes:

Use a scatterplot (hours studied → exam score). Draw the line: 'on average, more hours → higher score.' Emphasize: association ≠ causation.

# OLS in a sentence

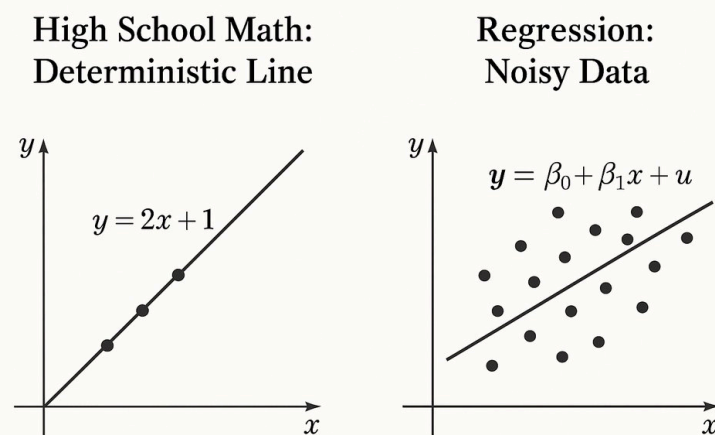
- Pick the line that makes residuals as small as possible (on average, squared)
- Use the line to predict and to test if the trend is basically flat

## Speaker Notes:

Residual = observed – fitted. OLS minimizes the sum of squared residuals. That's it.

# Variables vs Parameters – Why regression cares about $\beta$

- High-school line:  $y = 2x + 1$  is exact; every point lies on the line
- Regression world: data are noisy; we estimate fixed, unknown parameters  $\beta$
- Goal: learn  $\beta$  (population pattern), not chase each individual  $y$
- OLS picks the line that best summarizes the average relationship



Left: High-school line (deterministic) – every point lies on  $y = 2x + 1$ . Right: Regression with noisy data – estimate  $\beta$  from the cloud.

## Interpretation check

Wooldridge 7e – Chapter 2, Problems: Q2.1 (interpret slope/intercept)

In 'hours → score', suppose  $\beta_1 = 2.0$ . Explain this to a friend in plain English.

- Translate units: points per hour of study.
- Avoid causal language unless assumptions justify it.

**Answer:** On average, each extra hour of study is associated with about 2 more points on the exam.

## Rules

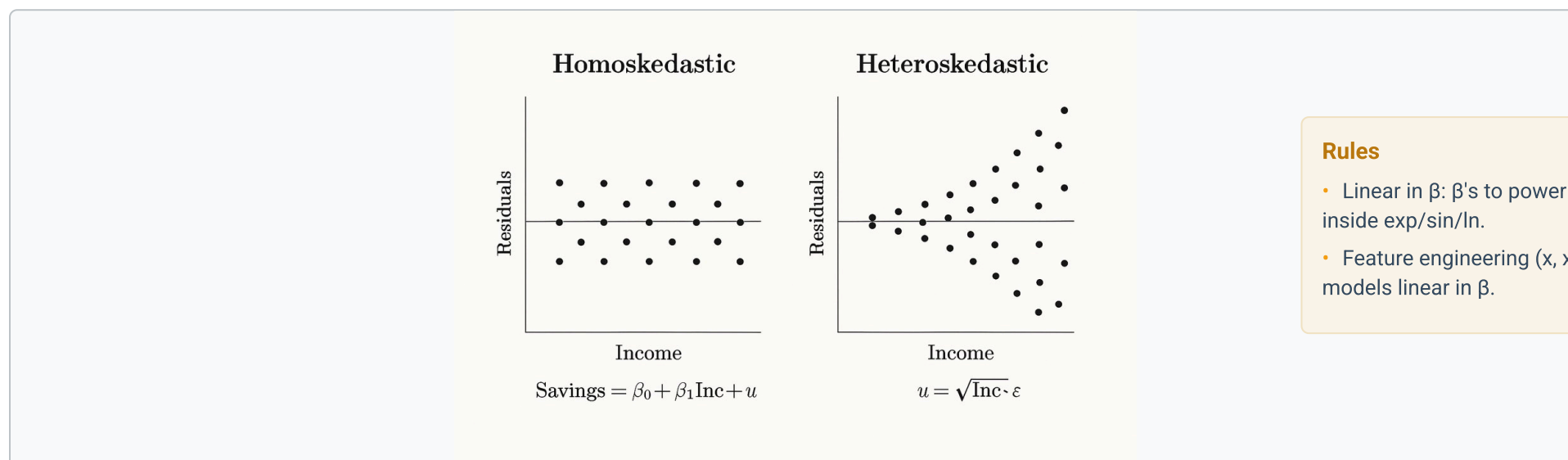
- We estimate  $\beta$ : fixed population parameters;  $x$  and  $y$  vary across samples.
- Residual = observed – fitted; OLS minimizes average squared residuals.

## Speaker Notes:

Contrast a deterministic function with a regression fit. In high school,  $y=2x+1$  means every  $(x,y)$  sits exactly on the line. In econometrics, we see a cloud of points. Our target is the fixed-but-unknown parameters  $\beta_0, \beta_1$  that describe the population's average relationship.  $x$  and  $y$  vary from sample to sample;  $\beta$  does not. This is why we estimate  $\beta$  and report uncertainty.

# Assumption 1: 'Linear' means linear in parameters ( $\beta$ ) – not necessarily a straight line in $x$

- Definition:  $\beta$ 's appear only to the first power; not multiplied or inside non-linear functions
- Curved in  $x$  can still be linear in  $\beta$  (e.g., add  $x^2$  as a feature)
- Examples (linear in  $\beta$ ):  $\beta_0 + \beta_1 x$ ;  $\beta_0 + \beta_1 x + \beta_2 x^2$ ;  $\beta_0 + \beta_1 \ln x$
- Non-examples (as written):  $\beta_0 + e^{\beta_1} x$ ;  $\beta_0 + \beta_1 \beta_2 x$ ;  $\beta_0 + \beta_1^2 x$



## Rules

- Linear in  $\beta$ :  $\beta$ 's to power 1; no  $\beta \times \beta$ ; no  $\beta$  inside exp/sin/ln.
- Feature engineering ( $x$ ,  $x^2$ ,  $\ln x$ ) can keep models linear in  $\beta$ .

Left: Linear in  $\beta$  ( $y = \beta_0 + \beta_1 x$ ). Middle: Still linear in  $\beta$  ( $y = \beta_0 + \beta_1 x + \beta_2 x^2$ ). Right: NOT linear in  $\beta$  ( $y = \beta_0 + e^{\beta_1} x$ ).

## Linear-in-parameters check

Wooldridge 7e – Chapter 2, Concept Check (in-class)

Which are linear in  $\beta$  as written? (1)  $y = \beta_0 + \beta_1 x^2$  (2)  $y = e^{\beta_0 + \beta_1 x}$  (3)  $y = \beta_0 + \beta_1^2 x$  (4)  $y = \beta_0 + \beta_1 \ln x + \beta_2 x$

- Rule:  $\beta$ 's to first power, added (not multiplied), not inside non-linear functions.
- Check each specification against the rule.

**Answer:** (1) Yes – linear in  $\beta$  ( $x^2$  is a feature). (2) No –  $\beta$ 's inside an exponential. (3) No –  $\beta_1$  enters squared. (4) Yes –  $\beta$ 's enter additively to the first power.

## Speaker Notes:

Common confusion: 'linear regression' is about linearity in  $\beta$ , not raw  $x$ . Quadratic regression  $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$  is still linear in  $\beta$  because  $\beta$ 's enter additively to the first power— $x^2$  is just another feature. This is why quadratic terms don't violate the linearity assumption.

# Assumption 2: Random sampling

- › Everyone came from the same fair process
- › Lets our sample mimic the population

## Bias spotter

You only survey students who attend the optional review. Random sample?

- Define the population.
- Ask if selection is related to y or x.

**Answer:** No — selection likely relates to hours and scores; conclusions may not generalize.

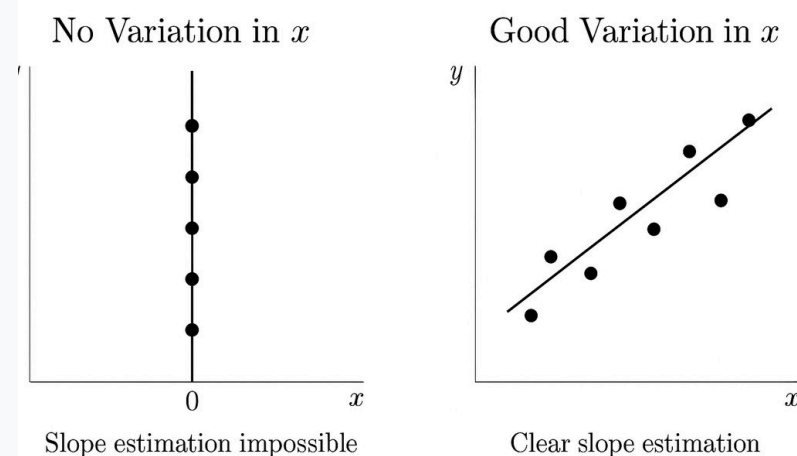
*Wooldridge 7e — Chapter 2, Problems: Q2.2/Q2.3 (sampling & interpretation)*

## Speaker Notes:

Non-random convenience samples can skew relationships.

# Assumption 3: Variation in $x$

- $x$  must vary to estimate a slope
- No variation  $\Rightarrow$  no information about  $\beta_1$



Left: No variation in  $x$  (all points at  $x=5$ ) makes slope estimation impossible. Right: With good variation in  $x$ , a clear slope can be estimated.

## Estimability check

Data: all students studied exactly 5 hours. Can we estimate  $\beta_1$ ?

- Slope needs differences in  $x$ .
- No variation  $\Rightarrow$  denominator = 0 in slope formula.

**Answer:** No — with no  $x$  variation,  $\beta_1$  cannot be estimated.

Wooldridge 7e — Chapter 2, Problems: Q2.4 (identification/variation)

### Rules

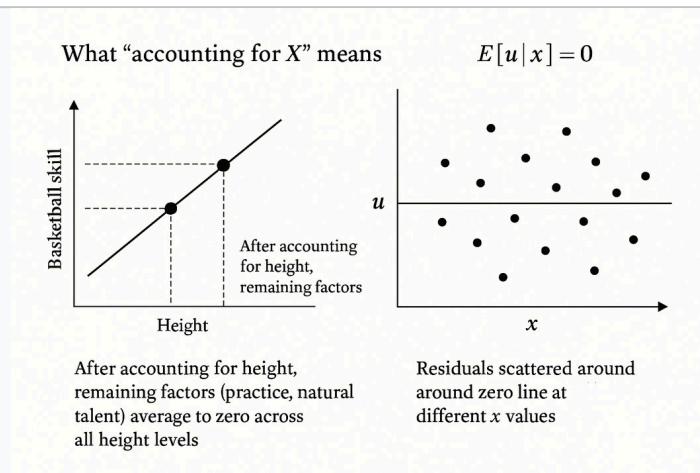
- Sample variance of  $x$  must be positive.
- Formula:  $\beta_1 = \text{Cov}(x,y)/\text{Var}(x) \rightarrow$  if  $\text{Var}(x)=0$ , then  $\beta_1$  is undefined.

## Speaker Notes:

If all  $x$  are the same, the best line is flat but  $\beta_1$  is not identified. Think about the formula for the OLS slope:  $\beta_1 = \sum(x_i - \bar{x})(y_i - \bar{y}) / \sum(x_i - \bar{x})^2$ . If all  $x$  values are identical, then every  $(x_i - \bar{x}) = 0$ , so the denominator = 0. You can draw a horizontal line at the mean of  $y$ , but you cannot identify any relationship between  $x$  and  $y$ .

# Assumption 4: Zero conditional mean ( $E[u|x]=0$ )

- After accounting for  $x$ , leftover noise isn't systematically high or low
- No big omitted factor tied to  $x$  (e.g., ability in wage~schooling)
- "Accounting for  $x$ " means after considering  $x$ 's effect, the average of all other influences is zero



Left: What "accounting for  $X$ " means in plain English. Right: Residuals should average to zero for each value of  $X$ .

## Savings Function Example

Wooldridge 7e – Chapter 2, Problem 7

Consider the savings function:  $\text{sav} = \beta_0 + \beta_1 \text{inc} + u$ , where  $u = \sqrt{\text{inc}} \cdot e$  and  $e$  is a random variable with  $E(e) = 0$ ,  $\text{Var}(e) = \sigma_e^2$ , and  $e$  is independent of  $\text{inc}$ . Show that  $E(u|\text{inc}) = 0$ .

- When we condition on  $\text{inc}$ ,  $\sqrt{\text{inc}}$  becomes a constant
- Use independence: if  $e$  is independent of  $\text{inc}$ , then  $E(e|\text{inc}) = E(e)$
- Apply the rule:  $E(u|\text{inc}) = E(\sqrt{\text{inc}} \cdot e|\text{inc}) = \sqrt{\text{inc}} \cdot E(e|\text{inc})$

### Rules

- LIE:  $E[Y] = E(E[Y|X])$
- $E[u|X]=0 \Rightarrow$  unbiased slope
- If  $e \perp X$ , then  $E(e|X) = E(e)$

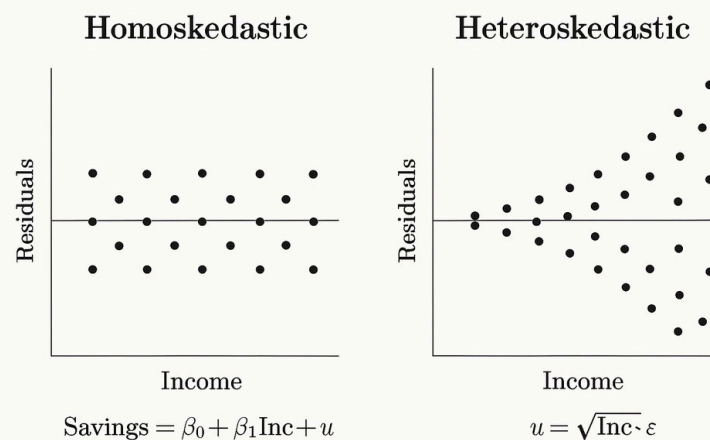
**Answer:**  $E(u|\text{inc}) = E(\sqrt{\text{inc}} \cdot e|\text{inc}) = \sqrt{\text{inc}} \cdot E(e|\text{inc}) = \sqrt{\text{inc}} \cdot E(e) = \sqrt{\text{inc}} \cdot 0 = 0$ . The zero conditional mean assumption is satisfied even though the error term depends on  $\text{inc}$ .

## Speaker Notes:

This is the 'no omitted variable bias' condition; it underpins unbiased slopes. The savings function example shows that even when the error term depends on  $X$  (through heteroskedasticity), the zero conditional mean can still hold if the underlying randomness is independent of  $X$ . This is a subtle but important distinction for understanding when our estimates will be unbiased.

# Assumption 5: Homoskedasticity

- Homoskedasticity: Noise has similar spread across all values of  $x$
- Heteroskedasticity: When variance of error depends on  $x$  (like  $u = \sqrt{\text{inc}} \cdot e$ )
- If variance increases with  $x$ , OLS still unbiased but standard errors need adjustment



## Rules

- $\text{Var}(u|x) = \sigma^2$  (constant) for all  $x$ : homoskedasticity
- $\text{Var}(u|x) = f(x)$ : heteroskedasticity
- Use robust SEs for heteroskedastic data

Left: Homoskedastic residuals with uniform variance across income levels. Right: Heteroskedastic residuals showing fan shape where variance increases with income ( $u = \sqrt{\text{inc}} \cdot e$ ).

## Savings function with heteroskedasticity

Wooldridge 7e – Appendix: Problem 7 (heteroskedasticity derivation)

Consider the savings function:  $\text{sav} = \beta_0 + \beta_1 \text{inc} + u$ , where  $u = \sqrt{\text{inc}} \cdot e$  and  $e$  is independent of  $\text{inc}$  with  $E(e) = 0$ ,  $\text{Var}(e) = \sigma_e^2$ . Show that  $\text{Var}(u|\text{inc}) = \sigma_e^2 \cdot \text{inc}$ , violating homoskedasticity.

- When conditioning on  $\text{inc}$ ,  $\sqrt{\text{inc}}$  becomes a constant
- Apply variance rules:  $\text{Var}(cX) = c^2 \text{Var}(X)$  where  $c$  is constant
- Use that  $\text{Var}(e|\text{inc}) = \text{Var}(e) = \sigma_e^2$  when  $e$  is independent of  $\text{inc}$

**Answer:**  $\text{Var}(u|\text{inc}) = \text{Var}(\sqrt{\text{inc}} \cdot e|\text{inc}) = (\sqrt{\text{inc}})^2 \text{Var}(e|\text{inc}) = \text{inc} \cdot \sigma_e^2$ . Since variance depends on  $\text{inc}$ , homoskedasticity is violated – variance of savings increases with income.

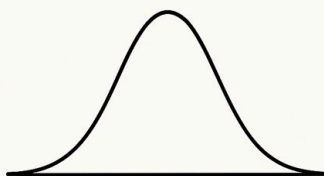
## Speaker Notes:

Heteroskedasticity is common in economics. Higher-income families have more discretion in spending vs. saving decisions, creating wider variability in savings behavior. Low-income families must spend most income on necessities (food, housing), leaving little room for savings variation. This explains why the variance of savings increases with income ( $\text{inc} \cdot \sigma_e^2$ ). While OLS remains unbiased under heteroskedasticity, traditional standard errors become unreliable, requiring robust standard errors for valid inference.

# Assumption 6: Normality (small samples)

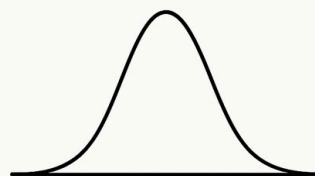
- Bell-curve errors make small-n t-tests exact
- With large n, CLT helps even if errors aren't perfectly Normal

Small Sample  
( $n = 40$ )



Normality matters more  
– exact t-tests rely on  
normal errors

Large Sample  
( $n = 400$ )



CLT helps  
approximate inference  
robust to non-normality

Left: Small sample ( $n=40$ ) - Normality matters more for exact t-tests. Right: Large sample ( $n=400$ ) - CLT makes inference robust to non-normality.

## When it matters

We have  $n=40$  vs  $n=400$ . When is Normality more critical?

- Small  $n$ : exact  $t$  relies on Normality.
- Large  $n$ : approximate by CLT.

**Answer:** At  $n=40$  it matters more; at  $n=400$ , CLT makes inference more robust.

*Wooldridge 7e – Chapter 2, Problems: Q2.7 (normality/use of t-tests)*

## Rules

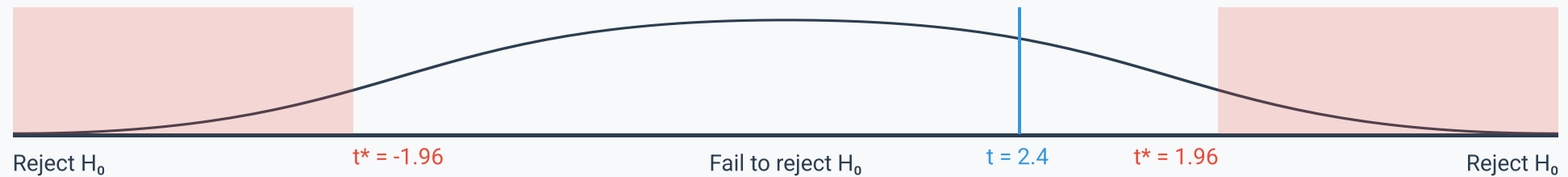
- Small  $n$ : exact  $t$  relies on Normality
- Large  $n$ : CLT approximation works well

## Speaker Notes:

Use for small classes; with big classes, focus on robust inference. Emphasize that Normality becomes less important with larger sample sizes thanks to the Central Limit Theorem, but remains crucial for precise inference with small samples.

# Testing the slope: is the line flat?

- $H_0: \beta_1 = 0$  (no linear trend)
- Test statistic = how many SEs away from 0



## Slope t-test (two-tailed)

Wooldridge 7e – Chapter 2, Problems: Q2.10 (testing  $\beta_1$ ) and Appendix C: Q.C6 (test logic)

Estimate:  $\hat{\beta}_1 = 2.4$  (points/hour),  $SE(\hat{\beta}_1) = 1.0$ ,  $n$  large. Test  $H_0: \beta_1 = 0$  at 5%.

- $t = \hat{\beta}_1 / SE = 2.4 / 1.0 = 2.4$ .
- Two-tailed  $p \approx 0.016$  (z-approx).
- Compare  $p$  to 0.05.

**Answer:**  $p \approx 0.016 < 0.05 \Rightarrow$  reject  $H_0$ . There's evidence of a positive linear trend in scores with hours.

### Rules

- $t = (\hat{\beta}_1 - 0) / SE(\hat{\beta}_1)$
- Reject if  $p \leq \alpha$  (or  $|t| \geq t^*$ ).

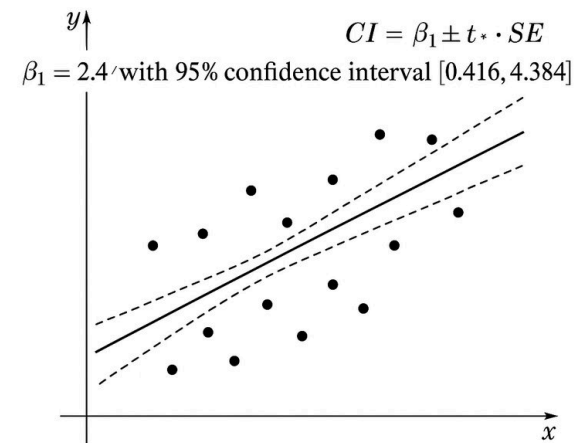
## Speaker Notes:

Explain p-value visually (tail area) and critical value rule; emphasize plain-English decisions.

# Confidence interval for the slope

- A range of slopes consistent with the data at 95%
- Report both estimate and uncertainty

Confidence Interval for Regression Slope



Point estimate  $\beta_1 = 2.4$  with 95% confidence interval  $[0.416, 4.384]$ . Formula:  $CI = \beta_1 \pm t^* \cdot SE$

## CI build

Using  $\beta_1 = 2.4$ ,  $SE = 1.0$ ,  $df \approx n - 2 = 98 \Rightarrow t^* \approx 1.984$  (95%). Construct the 95% CI.

- $CI = \beta_1 \pm t^* \cdot SE$
- Compute margins.

**Answer:**  $2.4 \pm 1.984 \times 1.0 \Rightarrow [0.416, 4.384]$ .

Wooldridge 7e – Chapter 2, Problems: Q2.11 (slope CI)

## Rules

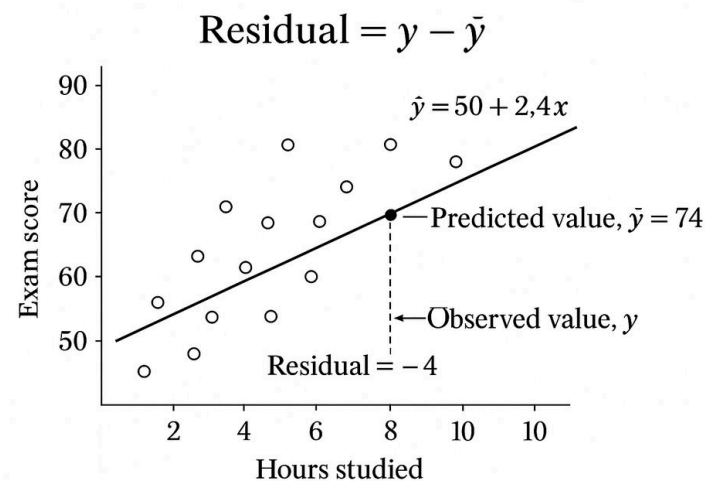
- CI: estimate  $\pm t^* \cdot SE$
- Interpret: 'plausible slopes at 95%'

## Speaker Notes:

CI communicates precision; readers can see practical significance.

# Predicting and interpreting

- Use the line to predict average  $y$  at  $x$
- Interpret in units; avoid extrapolation far from observed  $x$



Prediction and residual calculation: regression line  $\hat{y} = 50 + 2.4x$  showing prediction at  $x=10$  hours ( $\hat{y}=74$ ), actual score  $y=70$ , and the negative residual of  $-4$  points

## Fitted value & residual

Wooldridge 7e – Chapter 2, Problems: Q2.5 (fitted values/residuals)

$\beta_0=50$ ,  $\beta_1=2.4$ . (i) Predict score at  $x=10$ . (ii) If observed is 70, what's the residual? Interpret.

- (i)  $\hat{y} = 50 + 2.4 \cdot 10$ .
- (ii) Residual =  $y - \hat{y}$ .

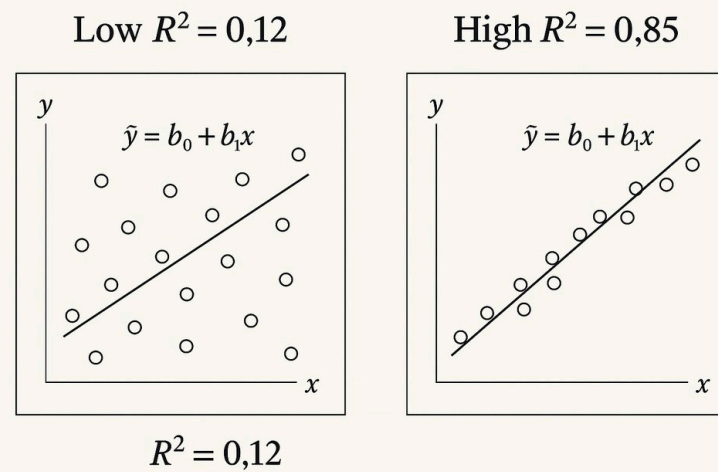
**Answer:** (i)  $\hat{y} = 74$ . (ii) Residual =  $70 - 74 = -4$ : the student scored 4 points below the model's prediction.

## Speaker Notes:

Connect prediction to decision-making (what you'd expect on average). Emphasize that fitted values represent our best guess for the average outcome at a specific value of  $x$ . Residuals tell us how far individual observations deviate from this average prediction. Remind students to be cautious about extrapolating far beyond the range of observed data as this can lead to unreliable predictions.

# Goodness-of-fit ( $R^2$ ) in one line

- $R^2$  = share of y-variation explained by the line
- High  $R^2 \neq$  causality; low  $R^2$  can still have a useful slope



Left: Low  $R^2$  (0.12) shows greater unexplained variation. Right: High  $R^2$  (0.85) shows data points following the regression line more closely.

## Meaning check

Wooldridge 7e – Chapter 2, Problems: Q2.6 ( $R^2$  interpretation)

$R^2=0.12$ . What does that tell you, and what does it not tell you?

- Fit vs causality.
- Slope could still be precisely estimated.

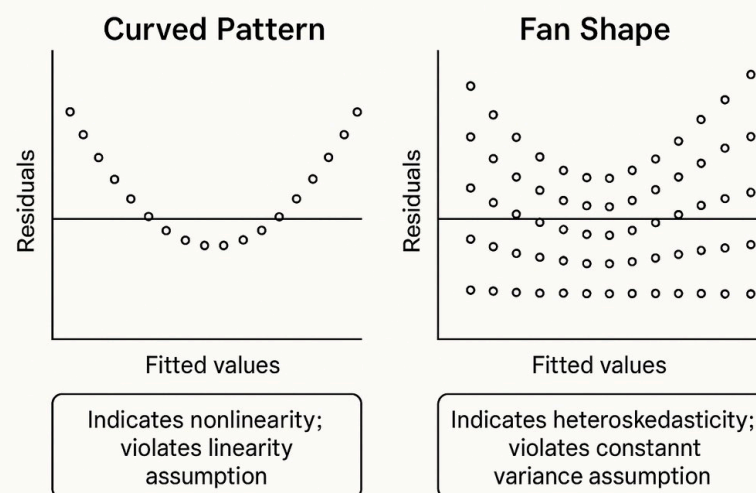
**Answer:** The line explains ~12% of score variation; it says nothing about causal impact. The slope can still be informative if estimated precisely.

## Speaker Notes:

Keep  $R^2$  in perspective: it's about fit, not identification.

# Quick diagnostics you can always do

- Residuals vs fitted: look for curve (nonlinearity) or fan (heteroskedasticity)
- Outliers/leverage: a single point can drag the line



Left: Curved pattern indicates nonlinearity (functional form issue). Right: Fan pattern indicates heteroskedasticity (non-constant error variance).

## Which assumption fails?

Wooldridge 7e – Chapter 2, Computer Exercises: C6/C7 (residual plots/outliers)

Your residual plot shows a curved pattern. Which assumption is most suspect and what could you try?

- Map symptom to assumption.
- Suggest a fix.

**Answer:** Linearity (functional form) looks violated. Consider adding a nonlinear term (e.g.,  $x^2$ ) or transforming variables.

## Speaker Notes:

One graph can prevent many bad conclusions; keep it routine. Residual plots are your first line of defense against model misspecification. A curved pattern in residuals vs. fitted values suggests the functional form is wrong. A fan-shape suggests heteroskedasticity. Always check for outliers with high leverage that could unduly influence your results.

# Wrap: what to remember

- Slope = average change; report with SE/CI
- Zero-conditional-mean is the linchpin
- Always sanity-check form & variance with a quick residual plot

## Speaker Notes:

Exit ticket: name the assumption you'd worry about most in  $\text{wage} \sim \text{schooling}$  and why.